Name:		Date:_________
Teacher:	Answers.	

Baldivis Secondary College	**Year 9 Mathematics** **Mini test 5** **Topic – Linear Equations** ***Full working out MUST be shown to get full marks for each question.***	SCORE: / 43

Total Time:	55 minutes	
Weighting:	5 %	
Equipment:	To be provided by the student: Pens and Pencils, scientific calculator, A4 page of notes.	

Part A – Multiple-choice questions **(5 marks)**

1. Which axis runs horizontally ?

A xy **B** y **C** x ✓
D z **E** Ø

2. How do you write an ordered pair ?

A (x , x) **B** (y , x) **C** (y , y) ✓
D (x , y) **E** (x , Ø)

3. What does 'c' represent in y = mx + c ?

A y - intercept **B** gradient **C** axis ✓
D x - intercept **E** coordinate

4. Write the equation of a line where the gradient is 0 and the y – intercept is 8.

A y = 8x **B** x = 8 **C** y = 8 ✓
D y = x + 8 **E** y = x + 8

5. What is the gradient in the picture?

A = positive **B** = negative **C** = zero **D** = Undefined ✓

Part B Short Answer Questions:

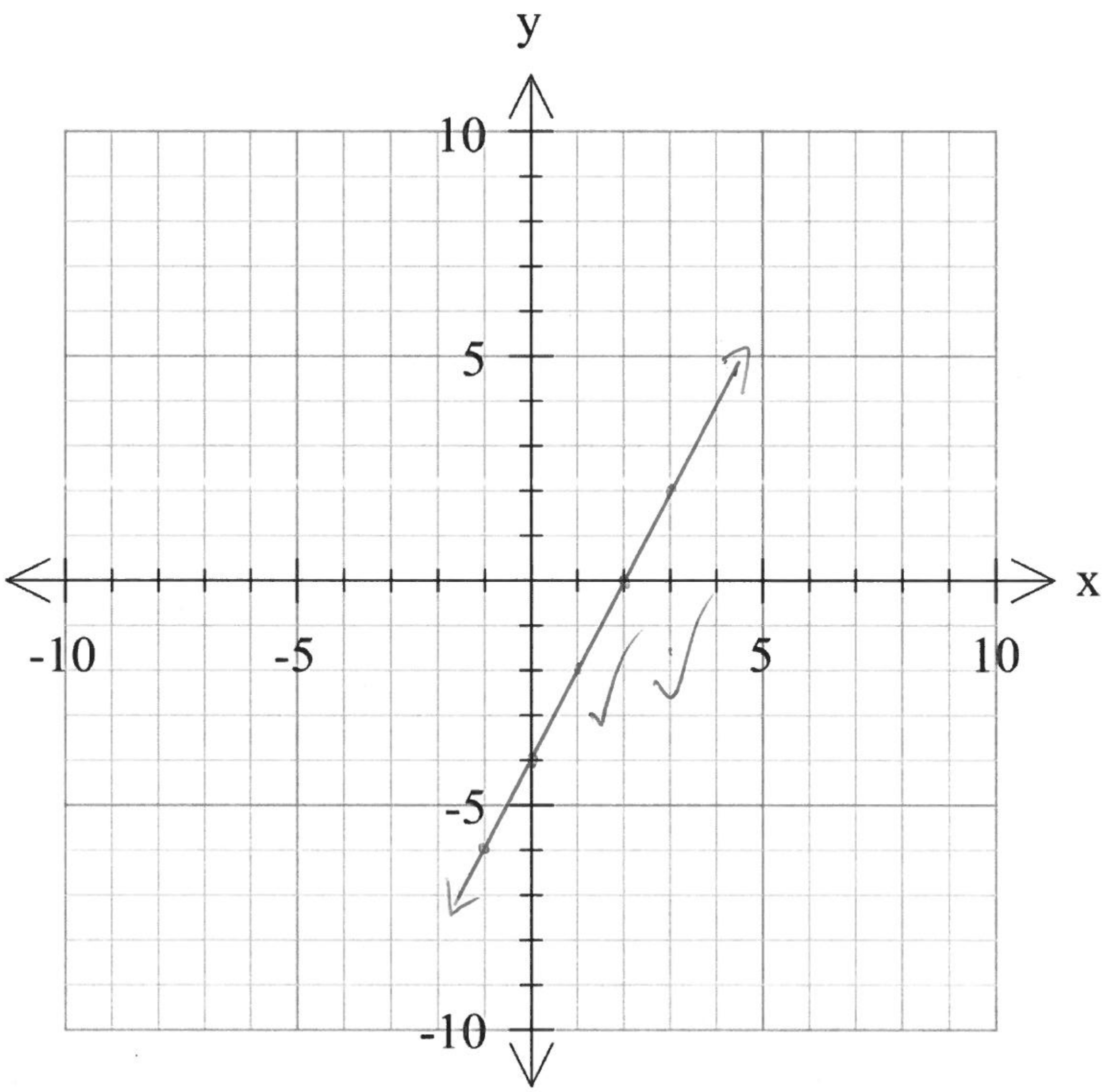
6. **(4, 2, 4 = 10 marks)**

(i) Complete the following tables of values from the rule below.

$$y = 2x - 4$$

x	-1	0	1	2	3
y	-6	-4	-2	0	2

(ii) Plot the coordinates from your table of values in (i) on the graph below and draw a straight line connecting the coordinates.



(iii) From the graph, write the **co-ordinates** of the following:

x-intercept:

(2, 0)

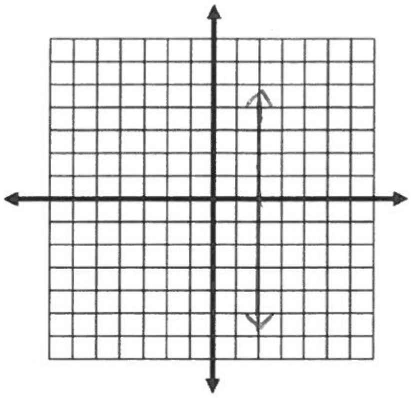
y-intercept:

(0, -4)

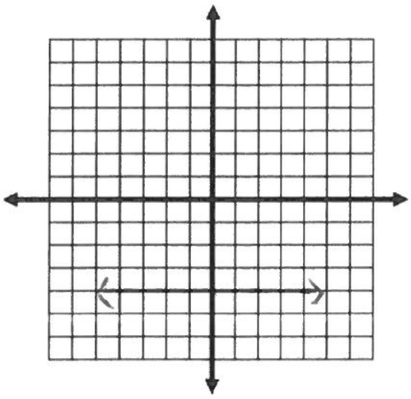
7. **(2, 2, 2, 2, 2 = 10 marks)**

Sketch the graphs of the following linear relationships (each small square = 1 unit).

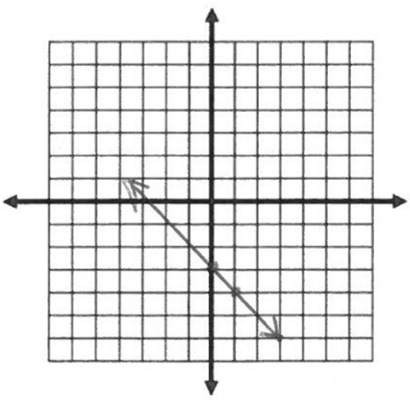
a. $x = 2$



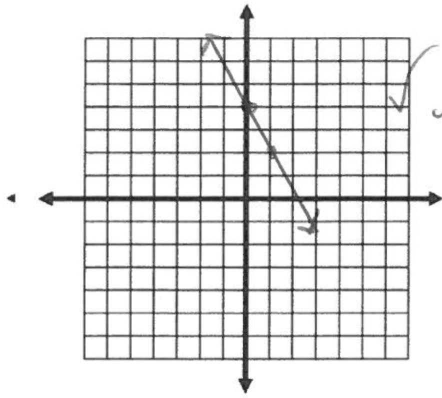
b. $y = -4$



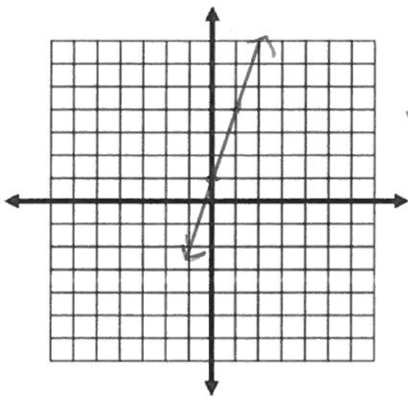
c. $y = -x - 3$



d. $y = -2x + 4$



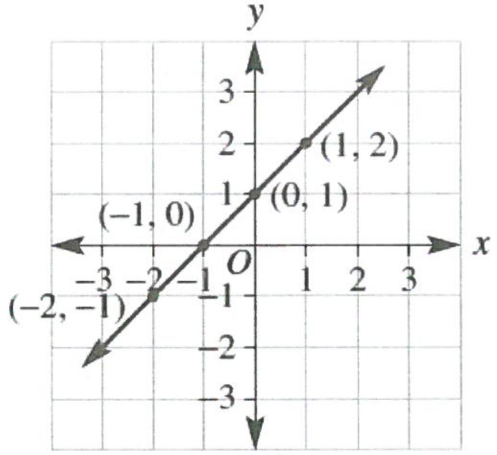
e. $y = 3x + 1$



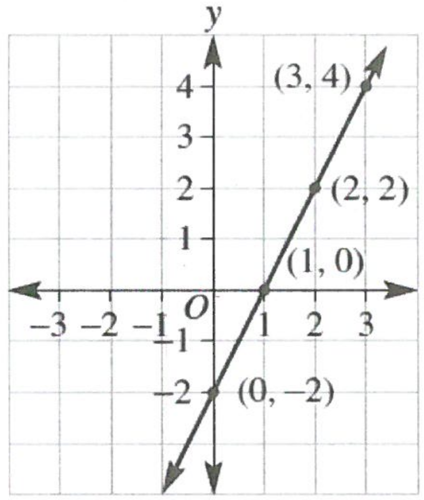
(−1 no arrows on line)

8. Match the following graphs with their correct gradients: **(4 marks)**

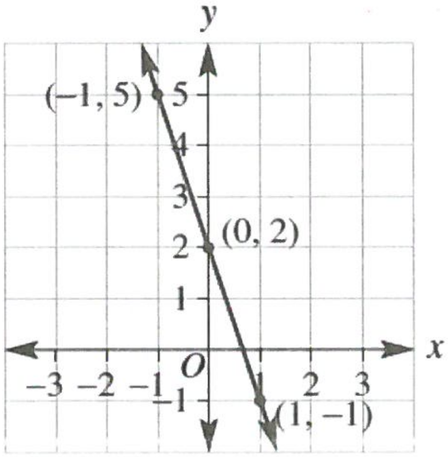
a



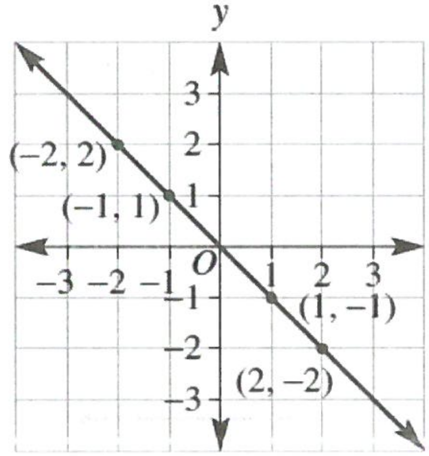
b



c



d



i. Gradient = -3: c ✓

ii. Gradient = 2: b ✓

iii) Gradient = 1: a. ✓

iv) Gradient = -1: d. ✓

9. 5 marks **(~~2 marks~~)**

Find the distance between and the midpoint between these two co-ordinates, (2,5) and (8, -4),

Midpoint.

$x - \frac{2+8}{2} = 5$ ✓

$y - \frac{5+(-4)}{2} = 0.5$

$(5, 0.5)$. ✓

Distance.

$x_2 - x_1 = 8 - 2 = 6$ units ✓

$y_2 - y_1 = -4 - 5 = 9$ units

$c = \sqrt{a^2+b^2}$

$= \sqrt{6^2+9^2}$ ✓

$= 10.82$ (2dp) ✓

Distance = 10.82 units.

10. **(4, 4, 2, 2 = 12 marks)**

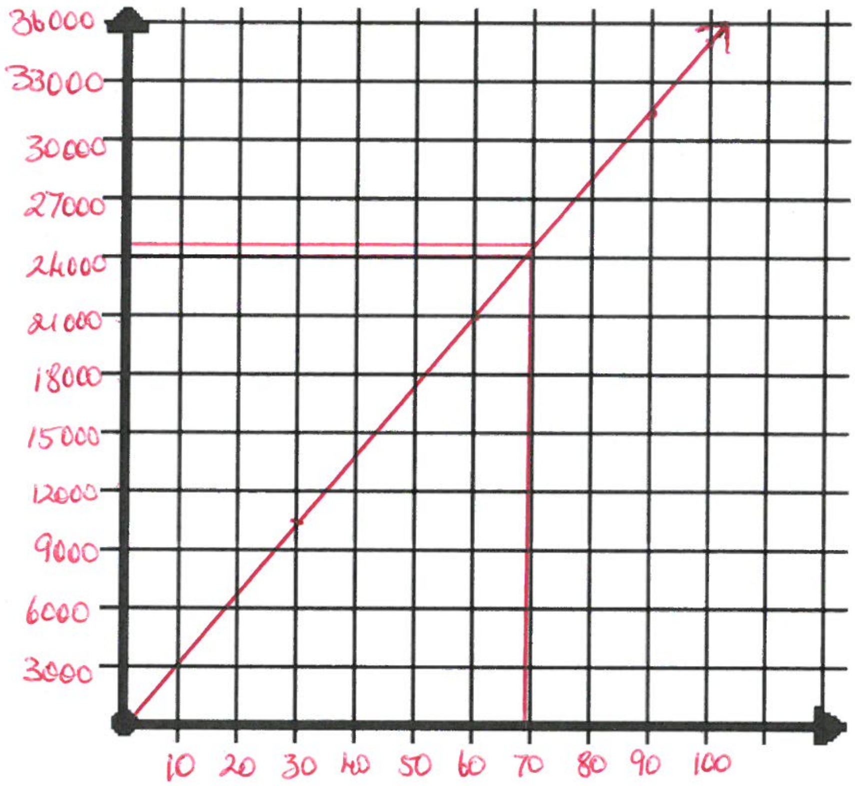
Phoebe's family decide to get a new pool and they ask her to calculate how long the pool will take to fill. Phoebe works out that the volume of water (v) in the pool is given by the equation $v = 350t$ where t is the time in hours.

a) Use the rule to complete the table of values.

t	0	30	60	90	100
v	0	10,500	21,000	31,500	35,000

b) Graph this relationship.

Volume of water vs time



c) If the pool is full after 70 hours, use your graph to estimate the volume of the pool (Show any working on the graph).

Estimated volume = 25000 L

1 mark for showing on graph.

d) Check your answer by substituting the relevant value into the equation. (Show your working)

Calculated volume = 24500 L

$V = 350(70)$
$= 24500$

(-1 No working)

End of test